Distributed Reinforcement Learning Based Market Mechanism Design for the Integration of Wholesale and Retail Electricity Markets

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***Abstract*—In this paper, we will consider the design of distributed optimization-based algorithmic market mechanisms and a unified system framework design, for decision making by both the traditional producers of electricity at the wholesale layer of the market and smart active consumers in addition to traditional consumers mostly at the retail layer, who can also feed power back to the grid. This unified framework for decision-making is based on a distributed model-based Reinforcement Learning (dm-RL), in which the environment is modeled as a Markov Decision Process (MDP) and Approximate Dynamic Programming (ADP) is used as the solution approach. For the the bulk power network, we will consider the post-contingency states representation and restoration through both generator redispatch and line-switching. For the consumers, we will employ the switching decisions as to which devices to connect/disconnect to the grid and also how much power to produce, consume, and curtail. The entire system will be an integrated framework of wholesale as well as the retail market layers. The decisions for both producers and consumers will pertain to solving for an optimal allocation of the amount of power to consume, produce, and curtail from several different sources. We will demonstrate the practical application of our ideas with a series of real world case-studies.**

***Index Terms*—Markov Decision Process (MDP), distributed model-based Reinforcement Learning (dm-RL), Approximate Dynamic Programming (ADP), Electricity Markets, Market Mechanism Design**

I. Notations and Conventions

We have categorized the entities used in the subsequent formulations into five different groups: Sets, Elements, Indices, Parameters, and Variables & Functions.

- *Sets*

  $\mathcal{N}$: Set of buses or nodes in the transmission or wholesale market layer
  $J(n)$: Set of nodes directly connected to node n
  D_n: Set of buses or nodes in the distribution or retail market layer, emanating from the nth node of the transmission system
  G: Set of generators
  H: Set of transmission lines
  $\mathcal{L} = \{0, 1, 2, ..., |\mathcal{L}|\}$: Set of possible $(N-1)$ contingencies. The element, 0 indicates the base case or healthy case, in which none of the transmission lines are outaged and they're all in service.

  Ω: Set of dispatch time intervals for which the receding horizon problem instances are solved

  $\dagger$ will be used to denote the transpose of a vector or matrix

- *Elements*

  n: Elements of $\mathcal{N}$
  $\overline{n}$: Elements of $J(n)$
  g: Elements of G
  h: Elements of H
  m: Elements of D_n
  c: Elements of $\mathcal{L}$
  $c \rightarrow c'$: List of contingency scenarios in the set $\mathcal{L} - \{c\}$, which will be used to denote, contingency scenarios, following the outage corresponding to scenario c
  τ: Elements of Ω which can also be interpreted as the "rolling" forthcoming dispatch interval

- *Indices*

  t: Future look-ahead dispatch time intervals
  d: Subscript to denote MW load (consumption)
  ν: Iteration count for the distributed optimization algorithmic market mechanism design

- *Parameters*

  $B_h^{(c)}$: Susceptance of the transmission line h in the contingency scenario (c)
  $\alpha_g, \beta_g, \gamma_g$: Quadratic, linear, and constant cost coefficients of the generator g
  $\overline{L}_h^{(0)}$: Long-term power carrying capacity of transmission line h
  $\overline{L}_h^{(c)}$: Short-term overload power carrying capacity of transmission line h in the contingency scenario (c)
  Γ_{RND}: Number of dispatch time intervals to restore the line flows such that they reach at below long-term rating, following a contingency
  Γ_{MRD}: Maximum number of allowed dispatch intervals, following an actual contingency to make the system secure with respect to the next set of contingencies
  $\beta^{(t)}$: Discount factor for dispatch time interval t
  $P_{dn}^{(t)}$: Aggregated load consumption at node n at time t (without any demand-response)
  $\pi_n^{(t)}$: Wholesale Locational Marginal Price (LMP) of electricity at node n at time t
  $\tilde{\pi}_n^{(t)}$: Retail price of electricity for all the leaf nodes and intermediate nodes in the distribution network connected to root transmission node n at time t
  $\overline{P}_g, \underline{P}_g, \overline{R}_g, \underline{R}_g$(=$-\overline{R}_g$, usually) denote the maximum and minimum generating limits & maximum ramp-up

and ramp-down limits of generators respectively.
ψ_{init} is the initial temperature of a transmission line
ψ_{amb} is the ambient temperature
α' is the fraction of line power that is the Ohmic loss (positive)
β' is the co-efficient of heat dissipation from the conductor due to convection and radiation (linearized).
$E_{\Gamma}^{(\omega)}$, $E_{t}^{(\omega)}$, $E_{\epsilon}^{(\omega)}$, $E_{0}^{(\omega)}$ are constants depending on the duration of the Γ_{RND}, nature of the conductor material, duration of a single dispatch interval, time of occurrence of the outage etc.
$\mathbf{\Phi}_{th}^{(c,\omega)} = \mathbf{\Phi}_{h}^{(c)} E_{t}^{(\omega)}, \forall t \in \{1, 2, ..., (\Gamma_{RND} - 1)\}$, $\forall \omega \in \{0, 1, 2, ..., (\Gamma_{RND} - 1)\}$,
$\mathbf{P}_{\text{inj}}^{(c)(\tau+t)} = (\mathbf{P}_{g}^{(c)(\tau+t)} - \mathbf{P}_{d}^{(\tau+t)}) \in \mathbb{R}^{|\mathcal{N}|}$ where each element of the vector is the injection at a particular node.
$\mathbf{\Phi}_{0h}^{(c,\omega)} = \mathbf{\Phi}_{h}^{(0)} E_{0}^{(\omega)} + \mathbf{\Phi}_{h}^{(c)} E_{\epsilon}^{(\omega)}$,
$\mathbf{\Phi}_{h}^{(c)} = [\Phi_{(h,1)}^{(c)}, ..., \Phi_{(h,|\mathcal{N}|)}^{(c)}]^{\dagger} [\Phi_{(h,1)}^{(c)}, ..., \Phi_{(h,|\mathcal{N}|)}^{(c)}]$.
where $\Phi_{(h,k)}^{(c)}$ refers to the element in the $(h,k)^{-th}$ position of the shift factor matrix of the network i.e. the change in power flow on the line h for unit MW injection at bus k and withdrawal at the system slack bus, corresponding to contingency scenario c. It is implicitly assumed that the above result is valid only for a preventive control scheme i.e. the case of DC-SCOPF with an $(N-1)$ contingency corresponding to line outage only.

- *Variables & Functions*

$P_{g}^{(t)}$: Power generation of generator g at time interval t
$\theta_{n}^{(t)}$: Bus voltage phase angle at node n at time interval t
$\tilde{W}_{m}^{(t)}$: Amount of load, connected to distribution node m to be curtailed at time t
$W_{m}^{(t)}$: Amount of power injected by the consumer connected to distribution node, m at time t
$X_{m}^{(t)}$: Amount of grid power withdrawn by the consumer connected to distribution node, m at time t
$Y_{m}^{(t)}$: Amount of power withdrawn by the consumer connected to distribution node, m at time t from solar panels
$Z_{m}^{(t)}$: Amount of power withdrawn by the consumer connected to distribution node, m at time t from rechargeable battery pack
$u_{mS}^{(t)}$: Switching decision for connecting the solar PV; 1 for connecting, 0 for not connecting
$u_{mZ}^{(t)}$: Switching decision for connecting the storage battery for either charging or discharging; 1 for connecting, 0 for not connecting
$u_{mX}^{(t)}$: Switching decision for connecting to the grid; 1 for connecting, 0 for not connecting
$u_{mW}^{(t)}$: Switching decision for feeding power to the grid; 1 for feeding power back to the grid, 0 for not
$u_{m\tilde{W}}^{(t)}$: Switching decision for curtailing demand; 1 for curtailing, 0 for not
$x_{h}^{(c)}$: Switching decision for keeping a transmission line h in operation, following outage corresponding to contingency scenario, c; 1 for keeping in operation, 0 for switching off.
$\mathbb{E}(.)$: Expectation operator
$U(.)$: Utility function of the consumer. $U(P_{dm} - \tilde{W}_{m}) = A_{d}(P_{dm} - \tilde{W}_{m})^{2} + B_{d}(P_{dm} - \tilde{W}_{m}) + K_{d}$ where $A_{d}, B_{d} \leq 0$ and $K_{d} \geq 0$. This means that for small amount of power, there's more utility for consuming power, rather than curtailing, whereas for large amounts, it's more beneficial to curtail some of the power, because typically a consumer would not want to consume more than a certain maximum. K_{d} is the positive utility in the form of a monetary payment to the consumer for just deciding to curtail it's demand.
$C_{g}(.)$: Cost function of conventional thermal generator $C_{g}(P_{g}) = \alpha_{g} P_{g}^{2} + \beta_{g} P_{g} + \gamma_{g}$

II. Introduction & Literature Survey

The traditional view of the electric power system, which is of controllable generation and probabilistic, but inflexible demand is changing, due to the inclusion of intermittent and probabilistic renewable generation, price-responsive smart loads, and new energy resources, such as plug-in electric vehicles, battery packs, consumer side solar PV etc. [1], [9], [11], [12], [23], [26], [33], [34]. As such, the modern day electric power system operation and planning are facing unprecedented paradigm shift driven by these factors. This means, that the generation and demand now share a lot of common features. While this shift of paradigm can potentially drive the system to the brink of its operating limits, it can also relieve the stress by providing enhanced preventive and corrective control capabilities to system failures. In this context, the problem of contingency modeling enhancements and need for developing a preventive-corrective dispatch scheme as emphasized by the CAISO in their straw proposals [13]–[22], deserve special mention. These are multi-dispatch interval OPF problems, for which the ISOs (especially CAISO) don't currently have a scheme, which represents not only the post-contingency restoration, while abiding by the thermal line limits, but also generates appropriate pricing signals to incentivize the different agents mentioned above to provide energy and reserves.

Another area in which this work generalizes some previous works and extends those, are the way we treat the consumer load. Our work in this paper, uses similar methodologies as of that presented in [25]. However, in that paper, the authors have assumed an inelastic, or a price insensitive load. In the present work, we will extend it to that of elastic load. In doing so, we will use the methodology of modeling the preferential switching of loads as a Markov chain, as described in [2]. Also, it is to be noted that, in [25], the authors have solved the decision making problem only for the consumer, whereas, in our work, we will solve the problem both for the consumer as well the producer. In fact, at the conclusion of this paper, we will see that the terms, producer and consumer have less significance. We will rather call them "prosumers," and we will have two types of prosumers, viz:

- High power prosumers (located in the transmission network or the meshed network part), and
- Low power prosumers (located in the distribution network or the tree network part)

As we can observe, from the foregoing discussion, since we are deriving an integrated framework for analyzing both the transmission as well as distribution and utilization layers of the electric network, we need the information pertaining to the topology of all the layers. The transmission layer topology is well known among the market participants usually. However, estimating the distribution layer topology is quite challenging. In the previous works [37], the authors have derived mapping rules between voltage, MW, and MVAr from partial measurements in the distribution network. The authors, in that paper used a Support Vector Machine (SVM) regression with a kernel trick, that allows for robust mapping rule derivation, in the absence of measurements that can render the complete observability of the network. A similar work, using a different methodology is [28], in which the authors have derived the topology of a distribution network having measurements for only a limited set of nodes. We point out, that in our work, we will have to solve the SCOPF for both the transmission and distribution network. Usually there is complete information and observability of the transmission network available with the system operator. However, the same cannot be said for the distribution network. Hence, we will be assuming that the methodologies proposed in [37] and [28] have been used to determine the mapping rule as well as the network topology, and we will use the pre-determined network topologies in our OPF formulation. In that sense, the present work extends the ideas presented in the above-mentioned couple references.

In summary, the following are the main goals of this paper:

- To explicitly solve the multi-stage dispatch problem, taking into account the "dynamic line rating", and producing market mechanisms for implementing demand response (DR), enhancing participation of renewable energy, smart loads etc (Solving this new SCOPF will automatically generate prices, which reflect need for reserves or additional resources to ensure system reliability).
- To maximize the penetration of renewable into the grid, like solar, wind, hydro etc. Can we have it as a component in objective in our optimization?
- To make use of flexible demands like EVs/PHEVs, battery charging, DRs, and load acting as reserves to attain system restoration and reliability, in case of unforeseen events as outages [4], [6], [10], [29], [32].
- To explore the possibilities of incentivizing the consumers to participate directly in the wholesale market layer through pricing schemes.

The entire scheme of this work is summarized in figure 1 below.

A. Technical Approach

In order to address each of the above-mentioned objectives, we will formulate the problem, in several stages, as mathematical optimization problems over multiple dispatch intervals, building in look-ahead capability. We will make use of the model predictive control paradigm, [3], [27], [31], [35], [36] since we want to build resiliency against suspected conditions in our formulation. Since the demand variation and renewable availability for the future are inherently probabilistic, we will adopt a Reinforcement Learning (RL)/Markov Decision Process (MDP)/Dynamic Programming (DP) approach to solve our optimization problem. Such problems become computationally expensive to solve. Hence we will make use of recent advances in distributed computations for power problems [5], [7], [24], [30] to solve our problem fast. Given below is the Bellman-Hamilton-Jacobi equation, which summarizes our approach

$$\begin{aligned} &V(\hat{S^{(t)}}, \mathbf{P}^{(t-1)}, \pi^{(t-1)}) \\ &= \max_{\mathbf{P}^{(t)} \in \mathcal{F}, \pi^{(t)}} \sum_{t=0}^{T} [NU(\mathbf{P}^{(t)} | \hat{S^{(t)}}) \\ &+ \sum_{S^{(t+1)} \in \mathcal{S}} p(S^{(t+1)} | \hat{S^{(t)}}) V(S^{(t+1)}, \mathbf{P}^{(t)}, \pi^{(t)})] \end{aligned} \tag{1a}$$

where,

- $V(\cdot,.,.)$ = Value function of state, $\hat{S^{(t)}}$ (associated with the state transitions)
- $p(.|.)$ = State transition probabilities
- $\mathcal{S}$ = State space of the system
- $NU(.)$ = Net utility function (Utility-Cost)
- $\pi^{(t)}$ = Vector of nodal prices for power at time slot t
- $S^{(t+1)}$ = Future state at time $t+1$
- $\hat{S^{(t)}}$ = Realization of current state
- $\mathcal{F}$ = Feasible set of power injections and other state variables, which obeys the constraints imposed by the Look-Ahead SCOPF problem
- $\mathbf{P}^{(t)}$ = Vector of power generation and consumption, which is composed of sub-vectors of conventional generation, wind, solar PV, solar concentrated, other renewable, conventional load, rechargeable batteries, PEVs/PHEVs, residential solar etc.

Observe, that equation (1) encapsulates the pricing incentives for motivating the loads to act as reserves and/or participate in the energy market by actively supplying power to the grid, in response to the pricing signals through the value function and an appropriate design of the utility function, based on which, the above model can even enhance the participation of smart/price responsive loads as well as renewable penetration. Figure 1 shows a schematic of the layers of the market, where we would also like to explore the possible underpinnings with and without the presence of the aggregator (and hence, it's dashed). The evaluation of the set $\mathcal{F}$ for a real-world system could be done by using algorithms like *Alternating Direction Method of Multipliers-Proximal Message Passing (ADMM-PMP)*, *Auxiliary Problem Principle (APP)* some of which has been done in the previous works [7], [8] or, by using other distributed algorithmic tools like, Optimality Condition Decomposition (OCD), Consensus+Innovation (C+I), Analytical Target Cascading (ATC) etc.

The specific tools that this work will produce will be the

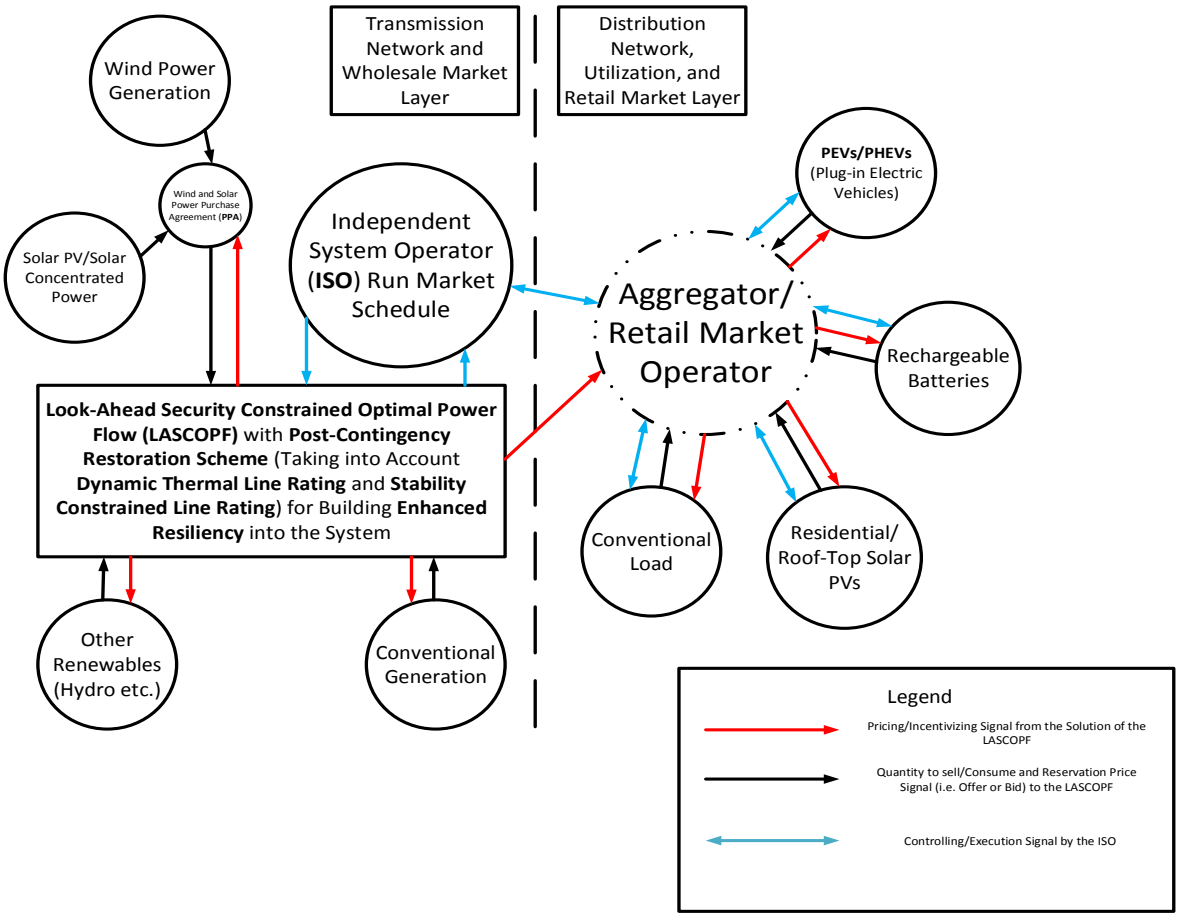

Fig. 1: Schematic Diagram of the Scheme of Integrating the Wholesale and Retail Market Layers

software simulation tool and the new multi-stage/multi-dispatch interval SCOPF, with built-in post contingency modeling and pricing scheme for incentivizing consumer participation and renewable penetration. These can be used as a decision making system.

III. Problem Description and Formulation

As laid out in the reference [25], we follow similar approach to state the problem. But, unlike that reference, here we state the optimization problem to be solved by both the producer and the consumer of electric power. The Expected Total Discounted Utility (ETDU) appearing in the objective function of the optimization problem (2) below is the net-utility to be maximized, shown previously in the Bellman-Hamilton-Jacobi equation (1)

$$\forall \tau \in \Omega_{hor}$$

$$\max_{\mathbf{P},\Theta,\pi,\mathbf{X}} \lim_{T\to\infty} \sum_{t=0}^{T} \beta^{(t)} \Bigg[\sum_{n\in\mathcal{N}} \Bigg\{ \sum_{g\in G_n} \mathbb{E}\Big(\pi_n^{(\tau+t)} P_g^{(\tau+t)} - C_g(P_g^{(\tau+t)}) \Big) + \sum_{m\in D_n} \mathbb{E}\Big(U(P_{dm}^{(\tau+t)} - \tilde{W}_m^{(\tau+t)}) + u_{m\tilde{W}}^{(t)} K_{dm} + \tilde{\pi}_n^{(\tau+t)}(W_m^{(\tau+t)} - X_m^{(\tau+t)}) + u_{mW}^{(t)} \tilde{\pi}_{min}^{(\tau+t)} \Big) \Bigg\} \Bigg] \tag{2a}$$

Subject to: (I) Producers' Constraints:

$$\forall \tau \in \Omega_{hor},\ \forall h \in H,\ \forall (c) \in \mathcal{L},\ \forall t \in \{0,1,2,...,\Gamma_{MRD}\}$$

$$\sum_{g\in G_n} P_g^{(0)(\tau)} - P_{dn}^{(\tau)} = \sum_{\overline{n}\in J(n)} B_h^{(0)}(\theta_n^{(0)(\tau)} - \theta_{\overline{n}}^{(0)(\tau)});\ \forall n \in \mathcal{N} \tag{2b}$$

$$\sum_{g\in G_n} P_g^{(0)(\tau)} - P_{dn}^{(\tau)} = \sum_{\overline{n}\in J(n)} B_h^{(c)}(\theta_n^{(c)(\tau)} - \theta_{\overline{n}}^{(c)(\tau)});\ \forall n \in \mathcal{N} \tag{2c}$$

$$\sum_{g\in G_n} P_g^{(c)(\tau+t)} - P_{dn}^{(\tau+t)} = \sum_{\overline{n}\in J(n)} x_h^{(c)} B_h^{(c)}(\theta_n^{(c)(\tau+t)} - \theta_{\overline{n}}^{(c)(\tau+t)});\ \forall n \in \mathcal{N} \tag{2d}$$

$$\sum_{g\in G_n} P_g^{(c)(\tau+\Gamma_{MRD})} - P_{dn}^{(\tau+\Gamma_{MRD})} = \sum_{\overline{n}\in J(n)} x_h^{(c)} B_h^{(c\to c')}(\theta_n^{(c\to c')(\tau+\Gamma_{MRD})} - \theta_{\overline{n}}^{(c\to c')(\tau+\Gamma_{MRD})});$$

$$\forall n \in \mathcal{N} \tag{2e}$$

$$|B_h^{(0)}(\theta_n^{(0)(\tau+t)} - \theta_{\overline{n}}^{(0)(\tau+t)})| \le \overline{L}_h^{(0)},\ \forall h \in H \tag{2f}$$

$$|B_h^{(c)}(\theta_n^{(c)(\tau+t)} - \theta_{\overline{n}}^{(c)(\tau+t)})| \le \overline{L}_h^{(c)},\ \forall h \in H \tag{2g}$$

$$\forall (c) \in \mathcal{L} - \{0\}, \forall (c') \in [\mathcal{L} - \{0, c\}],$$
$$\forall t \in \{0, 1, 2, ..., \Gamma_{MRD} - \Gamma_{RND}\}$$

$$|B_h^{(c)}(\theta_n^{(c)(\tau+\Gamma_{RND}+t)} - \theta_{\overline{n}}^{(c)(\tau+\Gamma_{RND}+t)})| \le x_h^{(c)}\overline{L}_h^{(0)},\ \forall h \in H \tag{2h}$$

$$|B_h^{(c\to c')}(\theta_n^{(c\to c')(\tau+\Gamma_{MRD})} - \theta_{\overline{n}}^{(c\to c')(\tau+\Gamma_{MRD})})| \le x_h^{(c)}\overline{L}_h^{(c\to c')},$$
$$\forall h \in H \tag{2i}$$

$$\forall \tau \in \Omega_{hor},\ \forall h \in H,\ \forall (c) \in \mathcal{L} - \{0\},\ \forall \omega \in \{0, 1, 2, ...(\Gamma_{RND} - 1)\}$$

$$E_\Gamma^{(\omega)}[\psi_{\text{init}}^{(\tau)}] + (1 - E_\Gamma^{(\omega)})[\psi_{\text{amb}}] + \left(\frac{\alpha'}{\beta'}\right)[\sum_{s=0}^{\Gamma_{RND}-(\omega+1)} (\mathbf{P}_{\text{inj}}^{(c)(\tau+t)\dagger} \mathbf{\Phi}_{sh}^{(c,\omega)} \mathbf{P}_{\text{inj}}^{(c)(\tau+t)})] < \psi_h^{\max} \tag{2j}$$

$$\underline{R}_g \le P_g^{(c)(\tau+1)} - P_g^{(0)(\tau)} \le \overline{R}_g,\ \forall g \in G \tag{2k}$$

$$\underline{R}_g \le P_g^{(c)(\tau+t+1)} - P_g^{(c)(\tau+t)} \le \overline{R}_g,\ \forall g \in G,\ (c) \in \mathcal{L},$$
$$t \in 1, 2, ..., \Gamma_{MRD} \tag{2l}$$

(II) Consumers' Constraints:

$$X_m^{(\tau+t)} + Y_m^{(\tau+t)} + Z_m^{(\tau+t)} \ge P_{dm}^{(\tau+t)} - \tilde{W}_m^{(\tau+t)} \tag{2m}$$

$$0 \le Y_m^{(\tau+t)} \le u_{mS}^{(\tau+t)} S_m^{(\tau+t)},\ \psi Z_m^{(\tau+t)} \le u_{mZ}^{(\tau+t)} c_{dis,m} \tag{2n}$$

$$-\psi \min\{Z_m^{(\tau+t)}, 0\} \le u_{mZ}^{(\tau+t)} c_{char,m} \tag{2o}$$

$$U_m^{(\tau+t+1)} - U_m^{(\tau+t)} = -\frac{\max\{Z_m^{(\tau+t)}, 0\}}{\eta_{dis,m}} - \eta_{char,m} \min\{Z_m^{(\tau+t)}, 0\} \tag{2p}$$

$$0 \le U_m^{(\tau+t)} \le K_m,\ u_{mX}^{(\tau+t)} X_m^{\min} \ge X_m^{(\tau+t)} \ge 0 \tag{2q}$$

(III) Aggregator's Constraints:

$$\sum_{g\in G} P_g^{(\tau+t)} + \sum_{n\in\mathcal{N}} \sum_{m\in D_n} W_m^{(\tau+t)} = \sum_{n\in\mathcal{N}} \sum_{m\in D_n} X_m^{(\tau+t)}, \forall t \tag{2r}$$

$$P_{dn}^{(\tau+t)} = \sum_{m\in D_n} (P_{dm}^{(\tau+t)} - \tilde{W}_m^{(\tau+t)}), \forall t, \forall n \in \mathcal{N} \tag{2s}$$

In the above optimization formulation of the problem, the first term in the objective function, (2a) represents the expected value of the generators' or traditional producers' surplus. It consists of the revenue from energy sale $\pi_n^{(\tau+t)} P_g^{(\tau+t)}$, which is the product of the LMP of the node to which a generator is connected and the MW output of the generator, less the cost of production, $C_g(P_g^{(\tau+t)})$. The second term, which represents the traditional consumers' expected surplus, is in turn composed of two parts, the first of which is the utility of the net consumed power $U_m(P_{dm}^{(\tau+t)} - \tilde{W}_m^{(\tau+t)})$ (the load without the demand-response $P_{dm}^{(\tau+t)}$, less the curtailed load due to the demand-response program $\tilde{W}_m^{(\tau+t)}$). The second part of the consumers' surplus $\tilde{\pi}_n^{(\tau+t)}(W_m^{(\tau+t)} - X_m^{(\tau+t)})$ is the net revenue, through the sale of power back into the grid $W_m^{(\tau+t)}$, less the amount of power withdrawn from the grid $X_m^{(\tau+t)}$. It is to be observed that the consumers' revenue is calculated based on the retail price of electricity $\tilde{\pi}_n^{(\tau+t)}$, rather than the wholesale price.

The constraints are classified into three major categories: Producers' Constraints, Consumers' Constraints, and Aggregator's Constraints.

Equalities (2b) and (2c) in the Producers' Constraints are the nodal power balance constraints for the base case and the contingency scenarios for the forthcoming dispatch interval. Each contingency scenario corresponds to the outage of one particular line among the N_c lines, considered for $(N-1)$ contingency analysis. Since, we are considering DC-OPF model, neglecting the line losses, there isn't any change of generation dispatch between healthy and contingency scenarios, although the bus voltage phase angles as well as the line susceptances change, as they should.

Equality constraint (2d) indicates nodal power balance for the look-ahead future dispatch intervals, on the assumption that any of the potential $(N-1)$ contingencies could have actually happened over the forthcoming dispatch interval and (2e) indicates the nodal power balance constraint corresponding to the fact that by the end of Γ_{MRD} number of dispatch intervals (which is the maximum allowed dispatch intervals to restore the system back to $(N-1)$ secure state, following an assumed outage), starting from the one after the forthcoming interval, the flows and voltage phase angles are such that the system is indeed restored to security with respect to the remaining contingencies.

Inequality constraints (2f) and (2g) represent the fact that the line flows are constrained to obey both the long-term thermal limits as well as the short term overload limits (for each contingency scenario) during the forthcoming dispatch interval and all the look -ahead dispatch intervals, under the assumption that the system still stays healthy by the end of the forthcoming dispatch interval. This simply assumes that if we are solving a look-ahead SCOPF, where we want to be secure with respect to any of the $(N-1)$ line contingencies at each interval, but there's no actual contingency happening in any of the intervals, under consideration, then what will be the optimal dispatch levels. The reason we have used the long-term thermal rating $L_h^{(0)}$ in inequality (2f) and the short term overload rating $L_h^{(c)}$ in (2g) is to indicate the fact that, as long as the system is healthy, the flows (which are less than long-term rating) are such that, if a line outage ever happens, then momentarily the flows on the remaining lines will exceed the long-term rating.

Constraints (2h) and (2i) play identical roles as that of constraints (2f) and (2g) in the situation, where it assumed that one of the contingencies might have actually happened during the forthcoming dispatch interval. However, now, it can be seen that, the flows on the remaining healthy lines are brought back to below the long term ratings, only after a delay of Γ_{RND} intervals, and the system is further secured against the remaining contingencies after Γ_{MRD} intervals. In

the time intervals from the forthcoming till Γ_{RND}, while the line flows are allowed to exceed the rated long-term values, the temperatures of the lines are monitored instead and constrained to not exceed the maximum allowed, according to constraint (2j). Constraints (2k) and (2l) denote the ramp rate limits for each generator. For the details of the derivation of these above-mentioned constraints and its solution, we refer the readers to [8], where the author proposes this problem, and uses a distributed algorithm composed of *Auxiliary Problem Principle (APP)* and *Alternating Direction Method of Multipliers-Proximal Message Passing (ADMM-PMP)* to solve it.

The consumers' constraints are the same as those that appear in [25]. For the sake of brevity, we will skip describing them here. The only difference is that the right hand side of (2m) is adjusted by the curtailment due to the demand response program. The aggregators' constraint (2r) makes sure that the total production, including power generated by generators, as well as that fed into the grid by consumers, equals the total power that is being drawn from the grid, while constraint (2s) ensures that total power consumed at each consumer node, after curtailment due to DR program, equals the demand at the respective load nodes of the transmission system. In the next section, we will describe the MDP/RL formulation of this complete problem.

IV. MDP/RL Modeling of the Problem

Following the same ideas that were introduced in [25], we will assume here, that, $\{P_{dn}^{(t)}\}_{t=0}^{\infty}$, $\{P_{dm}^{(t)}\}_{t=0}^{\infty}$, $\{S_m^t\}_{t=0}^{\infty}$, and $\{\tilde{\pi}_n^{(t)}\}_{t=0}^{\infty}$ are cyclic stochastic processes with cycle length, T hours (How is the cycle length determined?). Let $\Omega = \psi T$ denote the number of 5-minutes intervals in one cycle. Hence, if we write $k = t \bmod \Omega$, then $P_{dn}^{(t)} = \Lambda_{dn}^{(k)}\mathbb{P}_{dn}^{(t)} + \delta_{dn}^{(k)}$, $P_{dm}^{(t)} = \Lambda_{dm}^{(k)}\mathbb{P}_{dm}^{(t)} + \delta_{dm}^{(k)}$,$S_m^{(t)} = \xi_m^{(k)}\mathbb{S}_m^{(t)}$, $\tilde{\pi}_n^{(t)} = \Pi_n^{(k)}\mathcal{C}_n^{(t)} + \mu_n^{(k)}$. Here, $\{\Lambda_{dn}^{(1)}, \Lambda_{dn}^{(2)}, ..., \Lambda_{dn}^{(\Omega)}\}$, $\{\delta_{dn}^{(1)}, \delta_{dn}^{(2)}, ..., \delta_{dn}^{(\Omega)}\}$,$\{\Lambda_{dm}^{(1)}, \Lambda_{dm}^{(2)}, ..., \Lambda_{dm}^{(\Omega)}\}$,$\{\delta_{dm}^{(1)}, \delta_{dm}^{(2)}, ..., \delta_{dm}^{(\Omega)}\}$, $\{\xi_m^{(1)}, \xi_m^{(2)}, ..., \xi_m^{(\Omega)}\}$,$\{\Pi_n^{(1)}, \Pi_n^{(2)}, ..., \Pi_n^{(\Omega)}\}$,$\{\mu_n^{(1)}, \mu_n^{(2)}, ..., \mu_n^{(\Omega)}\}$ are constants (How are the values of these constants determined?) and $\{\mathbb{P}_{dn}^{(t)}\}_{t=0}^{\infty}$,$\{\mathbb{P}_{dm}^{(t)}\}_{t=0}^{\infty}$,$\{\mathbb{S}_m^{(t)}\}_{t=0}^{\infty}$, $\{\mathcal{C}_n^{(t)}\}_{t=0}^{\infty}$ are stationary independent Discrete Time Markov Chains (DTMCs). The state spaces for which are $\mathcal{S}_{dn}^{\mathbb{P}}$, $\mathcal{S}_{dm}^{\mathbb{P}}$, $\mathcal{S}_m^{\mathbb{S}}$, $\mathcal{S}_n^{\mathcal{C}}$ and the transition probability matrices are $\mathcal{P}_{dn}^{\mathbb{P}}$, $\mathcal{P}_{dm}^{\mathbb{P}}$, $\mathcal{P}_m^{\mathbb{S}}$, $\mathcal{P}_n^{\mathcal{C}}$ respectively. (The objective numerical estimation of the states and the probabilities in these needs to be determined through processing on big sets of measurement data, with the aid of several Machine Learning (and maybe Deep Learning) techniques?) Even though in reality, the actual DTMCs are continuous state-space, but we have implicitly assumed here, that we have discretized the state space into $M+1$ different values by dividing the entire range by a real number, M and considering the discrete steps as $\{0, 1/M, 2/M, ..., 1\}$ of the entire range.

The other stochasticity in the network, that we need to model, is the network topology for the transmission network, since we are considering the possibility of random outages. In order to model this factor, let's assume that we are going to consider N_c number of lines for $(N-1)$ contingency analysis. Each of those lines switches states between 0 and 1 with some probability. Hence, the state-space for the evolution of the network can be written as $\{0,1\}^{N_c}$. Therefore, $B_h^{(c)}$ for each look ahead time-step can assume states $B_h^{(c)} \neq 0$ and 0. Similarly, each $\overline{L}_h^{(0)}$ for each look-ahead dispatch interval can take values from the set $\{0, \overline{L}_h^l, \overline{L}_h^s\}$, where, the three values correspond to respectively the particular line in consideration being outaged, the system being healthy and no line being outages, and some other line except the one under consideration being outaged. Here $\overline{L}_h^l$ and $\overline{L}_h^s$ indicate the long-term and short-term thermal line limits respectively. Hence, the system state at time t can be written as the following 9-tuple:

$$\mathbf{x} = \{(t/\Omega)+1, \{\mathbb{P}_{dn}^{(t)}\}, \{\mathbb{P}_{dm}^{(t)}\}, \{\mathbb{S}_m^{(t)}\}, \{\mathcal{C}_n^{(t)}\}, \{\mathcal{U}_m^{(t)}\}, \{\overline{s}_h^{(0)}\}, \{B_h^{(c)N_c}\}, \{\overline{L}_h^{(0)}\}\} \tag{3a}$$

The DTMCs included within the $\{\}$ indicate a set of multiple DTMC, each of which in indexed by the corresponding subscripts. The state-space is then given by the Cartesian product between the following states:

$$\mathcal{S} = \{1, 2, ..., \Omega\} \times \mathcal{S}_{dn}^{\mathbb{P}|dn|} \times \mathcal{S}_{dm}^{\mathbb{P}|dm|} \times \mathcal{S}_m^{\mathbb{S}|m|} \times \mathcal{S}_n^{\mathcal{C}|n|} \times \mathcal{S}_m^{\mathcal{U}|m|} \times \{0,1\}^{N_c} \times \{0, B_h^{(c)}\}^{N_c} \times \{0, \overline{L}_h^l, \overline{L}s_h\}^{|H|} \tag{4a}$$

In the above description of the state-space, the superscript within the modulus sign denotes the fact that, each set is actually the Cartesian product of its component sets. For instance, $\mathcal{S}_{dn}^{\mathbb{P}|dn|}$ is the Cartesian product of $|dn|$ number of (which is the number of loads at the transmission level) state spaces, $\mathcal{S}_{dn}^{\mathbb{P}}$, each of which is the state-space of a particular load. The action at any time, t can be thought of as the amount of power withdrawn from the grid, the amount of demand curtailed, and the amount of power fed or sold to the grid, all by the consumers and the power generated by the generators. Therefore, the action space can be described as follows:

$$\mathcal{A}(\mathbf{x}) = \Big[\Big(P_{dm}^{(t/\Omega)} - S_m^{(t/\Omega)} - \min\Big\{\eta_{dis,m}U_m^{t/\Omega}, \frac{c_{dis,m}}{\psi}\Big\}\Big)^+, \Big(P_{dm}^{(t/\Omega)} - S_m^{(t/\Omega)} + \min\Big\{K_m - \frac{U_m^{t/\Omega}}{\eta_{char,m}}, \frac{c_{char,m}}{\psi}\Big\}\Big)^+\Big]^{|dm|} \times [0, P_{dm}^{t/\Omega}]^{|dm|} \times \Big[\Big(S_m^{(t/\Omega)} - \min\Big\{K_m - \frac{U_m^{t/\Omega}}{\eta_{char,m}}, \frac{c_{char,m}}{\psi}\Big\}\Big)^+, \Big(S_m^{(t/\Omega)} + \min\Big\{\eta_{dis,m}U_m^{t/\Omega}, \frac{c_{dis,m}}{\psi}\Big\}\Big)^+\Big]^{|dm|} \times [\underline{P}_g, \overline{P}_g]^{|G|} \tag{5a}$$

In the above, $(.)^+ = \max\{., 0\}$. The lower bound of the action space for the amount of power withdrawn from the grid (the first term above) is that needed to fulfill the demand when the battery is greedily discharged, while the upper bound for the same is the one corresponding to meeting the demand and greedily charging the storage. The second term, which corresponds to the amount of demand to be curtailed, varies between zero and the maximum demand at any point of time. The third term is for the power fed into the grid, and it can be observed that, it's very similar to the power drawn from the grid (without the terms for demand), but the limits are reversed, as is obvious. Lastly, the fourth term is for the generators'

decision. The "stage-cost" at time, t will be the total cost for both types of prosumers, the traditional producers and the traditional consumers. (The calculation of overall Transition probabilities from the individual transition probabilities, as mentioned above, through Kronecker products, for systems of real world scale, need Machine learning techniques???) The MDP problem for the overall problem can be stated as follows:

$$\max_{J_{\mathbf{x}}} \sum_{\mathbf{x}\in\mathcal{S}} c_{\mathbf{x}} J_{\mathbf{x}} \tag{6a}$$

$$\text{Subject to: } g_{\mathbf{x}}(a) + \beta \sum_{\mathbf{y}\in\mathcal{S}} P_{\mathbf{xy}}(a) J_{\mathbf{y}} \geq J_{\mathbf{x}}$$

$$\forall \mathbf{x} \in \mathcal{S}, \forall a \in \mathcal{A}(\mathbf{x}) \tag{6b}$$

In the above, $g_{\mathbf{x}}(a)$ is the stage cost, $J_{\mathbf{x}}$ is the cost-to-go at state $\mathbf{x}$ and a is the action taken. $\{c_{\mathbf{x}}\}_{\mathbf{x}\in\mathcal{S}}$ is a vector of positive values, which is given. The optimal action to be taken at each state is obtained by solving the Bellman equation as follows:

$$a^* = \text{argmin}_{a\in\mathcal{A}(\mathbf{x})} \Big\{ g_{\mathbf{x}}(a) + \beta \sum_{\mathbf{y}\in\mathcal{S}} P_{\mathbf{xy}}(a) J^*_{\mathbf{y}} \Big\} \tag{7a}$$

V. Distributed Optimization Algorithmic Mechanism Design

In this section, we will present an algorithmic mechanism design, which will split the entire optimization problem (2), into separate sub-problems for producers, consumers, and aggregators. The action variables, are the decision variables and at any iteration ν of the algorithmic mechanism design, will be denoted with superscripts with the iteration count. The aggregators' problems are as follows:

$$\sum_{g\in G} P_g^{(\tau+t)(\nu)} + \sum_{n\in\mathcal{N}} \sum_{m\in D_n} W_m^{(\tau+t)(\nu)} = \sum_{n\in\mathcal{N}} \sum_{m\in D_n} X_m^{(\tau+t)(\nu)}$$
$$\leftrightarrow \mu^{(\tau+t)(\nu)}, \forall t \tag{8a}$$

$$P_{dn}^{(\tau+t)} = \sum_{m\in D_n} (P_{dm}^{(\tau+t)} - \tilde{W}_m^{(\tau+t)(\nu)})$$
$$\leftrightarrow \lambda_n^{(\tau+t)(\nu)}, \forall t, \forall n \in \mathcal{N} \tag{8b}$$

The quantities on the right hand side of the double headed arrow are the Lagrange multipliers. These Lagrange multipliers can be interpreted as penalties for failing to satisfy the constraints (9a) and (9b), and are updated as follows:

$$\mu^{(\tau+t)(\nu+1)} = \mu^{(\tau+t)(\nu)} + \alpha_\mu \Big(\sum_{g\in G} P_g^{(\tau+t)(\nu)}$$
$$+ \sum_{n\in\mathcal{N}} \sum_{m\in D_n} W_m^{(\tau+t)(\nu)} - \sum_{n\in\mathcal{N}} \sum_{m\in D_n} X_m^{(\tau+t)(\nu)} \Big), \forall t \tag{9a}$$

$$\lambda_m^{(\tau+t)(\nu+1)} = \lambda_m^{(\tau+t)(\nu)} + \beta_\lambda \Big(P_{dn}^{(\tau+t)} - \sum_{m\in D_n} (P_{dm}^{(\tau+t)}$$
$$- \tilde{W}_m^{(\tau+t)(\nu)}) \Big), \forall t, \forall n \in \mathcal{N} \tag{9b}$$

The new objective function now can be written as

$$\max_{\mathbf{P},\Theta,\pi,\mathbf{X}} \lim_{T\to\infty} \sum_{t=0}^{T} \beta^{(t)} \mathbb{E} \Bigg[\sum_{n\in\mathcal{N}} \Bigg\{ \sum_{g\in G_n} \Big(\pi_n^{(\tau+t)} P_g^{(\tau+t)}$$
$$- C_g(P_g^{(\tau+t)}) \Big) - \mu^{(\tau+t)(\nu)} \Big(\sum_{g\in G_n} P_g^{(\tau+t)}$$
$$+ \sum_{m\in D_n} W_m^{(\tau+t)} - \sum_{m\in D_n} X_m^{(\tau+t)} \Big)$$
$$+ \sum_{m\in D_n} \Big(U(P_{dm}^{(\tau+t)} - \tilde{W}_m^{(\tau+t)})$$
$$+ \tilde{\pi}_n^{(\tau+t)} (W_m^{(\tau+t)} - X_m^{(\tau+t)})$$
$$- \lambda_m^{(\tau+t)(\nu)} \Big(P_{dn}^{(\tau+t)} - (P_{dm}^{(\tau+t)} - \tilde{W}_m^{(\tau+t)}) \Big) \Big) \Bigg\} \Bigg] \tag{10a}$$

Hence, the entire optimization problem can now be split into these different ones. The one given below are for producers:

$$P_g^{(\tau+t)(\nu+1)}$$
$$= \text{argmax}_{\mathbf{P},\Theta,\pi,\mathbf{X}} \lim_{T\to\infty} \sum_{t=0}^{T} \beta^{(t)} \mathbb{E} \Bigg[\sum_{n\in\mathcal{N}} \Bigg\{ \sum_{g\in G_n} \Big(\pi_n^{(\tau+t)} P_g^{(\tau+t)}$$
$$- C_g(P_g^{(\tau+t)}) \Big) - \mu^{(\tau+t)(\nu)} \sum_{g\in G_n} P_g^{(\tau+t)} \Bigg\} \Bigg] \tag{11a}$$

$$\text{Subject to: } (2b) - (2l) \tag{11b}$$

The following ones for each consumer.

$$\forall n \in N, \forall m \in D_n, W_m^{(\tau+t)(\nu+1)}, X_m^{(\tau+t)(\nu+1)}, \tilde{W}_m^{(\tau+t)(\nu+1)}$$
$$= \text{argmax}_{\mathbf{P},\Theta,\pi,\mathbf{X}} \lim_{T\to\infty} \sum_{t=0}^{T} \beta^{(t)} \mathbb{E} \Bigg[- \mu^{(\tau+t)(\nu)} \Big(W_m^{(\tau+t)} - X_m^{(\tau+t)} \Big)$$
$$+ \Big(U(P_{dm}^{(\tau+t)} - \tilde{W}_m^{(\tau+t)})$$
$$+ \tilde{\pi}_n^{(\tau+t)} (W_m^{(\tau+t)} - X_m^{(\tau+t)})$$
$$- \lambda_m^{(\tau+t)(\nu)} \Big(P_{dn}^{(\tau+t)} - (P_{dm}^{(\tau+t)} - \tilde{W}_m^{(\tau+t)}) \Big) \Big) \Bigg\} \Bigg] \tag{12a}$$

$$\text{Subject to: } (2m) - (2q) \tag{12b}$$

Optimization problems (11) and (12) along with (9) form a distributed algorithmic market mechanism design, in which, each consumer and the producers update their own decisions, after receiving the penalty signal from the aggregator in the form of updated Lagrange multipliers. The aggregator gathers the updated decision variables after each iteration and updates the Lagrange multipliers, and then broadcasts those. until the convergence criteria is met.

VI. Algorithms for Solving the MDP

In order to numerically solve the RL/MDP problem for market integration, we will use the **Approximate Dynamic Programming (ADP)** and two of its variants, namely, the **One Step Look-ahead Algorithm (OLA)** and **One Step Roll-out Algorithm (ORA)** as mentioned in [25]. For the OLA, given the current system state, $\mathbf{x}$, the action that is chosen corresponds to the optimal values of the decision variables in optimization problems (9)-(12), but with $T = 1$. For the ORA, for a given number of future time intervals or time steps, in order to

Fig. 2: The 5 Bus Power System.

calculate the cost to go, we calculate the cost to go by solving the optimization problems (9)-(12), with the stochastic state parameters, replaced by the corresponding conditional expected values, conditioned over a particular realization of the state variables, and then solving the Bellman equation of minimizing the sum of the stage cost and the discounted expected cost to go, over the entire state-space.

VII. Numerical Simulations

We will present here case studies for some systems we have run using PowerLASCOF.jl and PowerData.jl. These two are open-source power system operations simulation model and power system data model, respectively, written in the Julia programming language with JuMP optimization modeling framework. Both of these models are built up around the existing ecosystem of Julia-based models, namely: National Renewable Energy Laboratory (NREL)'s Sienna suite (PowerSystems.jl, PowerSimulations.jl, InfrastructureSystems.jl, PowerModelsInterface.jl etc.), Los Alamos National Laboratory-Advanced Network Science Initiative (LANL-ANSI)'s PowerModels.jl and InfrastructureModels.jl, Lawrence Livermore National Laboratory (LLNL)'s Egret (written in Python), and Princeton University-ZERO Lab+Massachusetts Institute of Technology-Energy Initiative (PU+MITEI)+New York University (NYU) Tandon School's GenX. The APMP nested decomposition algorithm (consisting of two outer APP layers for decomposition across dispatch time intervals and contingency scenarios and an inner ADMM-PMP layer for decomposition across devices and nodes) has been implemented utilizing a graph-theoretic structure and optimization using Plasmo.jl. In this section, we present the results of the APMP algorithm for the LASCOPF problem instance for tracking demand variation for the 5 bus system. All the other LASCOPF problems can be solved in the exact same manner.

TABLE I: Variation of Load for 5 Bus System.

Conn. Node	MW, Int.-1	MW, Int,-2	MW, Int.-3
2	20	30	20
3	45	40	43
4	40	40	45
5	60	65	65

TABLE II: Generator Outputs 5 Bus System.

Generator	MW, Int.-1	MW, Int.-2	MW, Int.-3
Gen. 1 (Conn. Node: 1)	140.771	141.355	143.131
Gen. 2 (Conn. Node: 2)	24.229	33.6459	29.8704

The figure 3 shows the outer APP convergence for this example problem.

Residuals.pdf

APP Residuals.pdf

Fig. 3: APP Convergence Characteristics of 5 Bus System LASCOPF

A. APP Convergence Characteristics

All the rsults presented above have been calculated using a single thread, although, the code is written in such a way, that, if desired, it can easily be converted to a multithreaded one. Implementing fully multithreaded code is one of the proposed future works.

VIII. Conclusion

In this paper, we have presented the design of a market mechanism for building an integrated framework for the decision making in the electricity market. Unlike the traditional approach, in our paper, we have considered both the wholesale market layer producers and the end consumers as having the capability of decision making and feeding power to the grid, as prosumers. Our market mechanism design is based on a distributed Reinforcement Learning (RL)/Markov Decision Process (MDP) algorithm. We have considered the post-contingency states representation and restoration as well as price-responsive flexible loads. We have illustrated the mechanism through some simulation results on the IEEE test systems. As the future steps, we intend to extend this mechanism to AC-OPF and incorporating the distribution network topology explicitly.

Acknowledgement: The authors would like to extend their gratitude to Yang Weng, Professor, Arizona State University; Feng Pan and Henry Huang, both with the Pacific Northwest National Laboratory (PNNL), Richland, WA, for their assistance in formulating several crucial parts of this paper and also for their guidance.

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